

What Are URL Converters in Flask?

Definition:

URL converters are used **inside a route** (in your Flask code) to tell Flask **what kind of data type** is expected from the URL.

They **extract and convert** parts of the URL into variables automatically.

Basic Syntax

```
@app.route('/route/<converter:variable>')
```

```
def function_name(variable):
```

```
    return ...
```

Here:

- **converter** → tells Flask what type of data to expect (like int, float, string...)
 - **variable** → name of the variable passed into your function
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Flask Built-in URL Converters

Converter	Description	Example URL	Converted Data Type
string	Default type; accepts text (no slash /)	/user/jani	'jani' (string)
int	Accepts only integers	/user/10	10 (int)
float	Accepts only decimal numbers	/value/4.5	4.5 (float)
path	Accepts text with slashes /	/file/images/photo.jpg	'images/photo.jpg' (string)
uuid	Accepts UUID strings	/id/123e4567-e89b-12d3-a456-426614174000	UUID object

♦ string (Default Converter)

 **Used for:** Simple words (no slash /)

If no converter is specified, string is used by default.

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route('/user/<string:name>')
```

```
def show_user(name):
```

```
    return f"Hello, {name}!"
```

 URL:

`http://localhost:5000/user/Jani`

➡ **Output:**

Hello, Jani!

 If you try `/user/jani/basha`, you'll get **404 Not Found** (because of `/`).

♦ **int (Integer Converter)**

 **Used for:** Numbers only (no decimal point)

If you pass text, Flask gives **404**.

```
@app.route('/square/<int:num>')
```

```
def square(num):
```

```
    return f"Square of {num} = {num ** 2}"
```

 URL:

`http://localhost:5000/square/5`

➡ **Output:**

Square of 5 = 25

 `/square/abc` → 404 Error

 `/square/5` → Works fine

♦ **float (Decimal Number Converter)**

 **Used for:** Decimal (floating-point) numbers

```
@app.route('/half/<float:value>')
```

```
def half(value):
```

```
    return f"Half of {value} is {value / 2}"
```

 URL:

http://localhost:5000/half/8.4

➡ **Output:**

Half of 8.4 is 4.2

 /half/hello → 404 Not Found

♦ **path (Path Converter)**

 **Used for:** Strings **that can include slashes (/)**

Useful for file paths, folder names, etc.

```
@app.route('/file/<path:filename>')
```

```
def show_file(filename):
```

```
    return f"File path: {filename}"
```

 URLs & Outputs:

URL	Output
/file/photo.jpg	File path: photo.jpg
/file/images/photo.jpg	File path: images/photo.jpg
/file/docs/work/report.pdf	File path: docs/work/report.pdf

♦ **uuid (Universal Unique Identifier Converter)**

 **Used for:** UUID values (standard unique 128-bit strings)

Flask automatically converts them into UUID objects.

```
import uuid
```

```
@app.route('/item/<uuid:item_id>')
```

```
def get_item(item_id):
```

```
    return f"Item ID is {item_id}"
```

 URL:

http://localhost:5000/item/123e4567-e89b-12d3-a456-426614174000

➡ **Output:**

Item ID is 123e4567-e89b-12d3-a456-426614174000

✗ /item/abc → 404 Error

🧠 Summary Table

Converter	Accepts	Example URL	Example Output	Use Case
string	Text (no /)	/user/jani	"jani"	Names, words
int	Whole numbers	/age/21	21	IDs, counts
float	Decimal numbers	/price/45.6	45.6	Prices, percentages
path	Text with /	/file/images/photo.jpg	"images/photo.jpg"	File paths
uuid	UUID strings	/data/123e4567-e89b-12d3-a456-426614174000	UUID object	Unique IDs

🔹 6 (Optional) Custom Converters

You can even **create your own** converter — for example, to allow only even numbers.

from werkzeug.routing import BaseConverter

```
class EvenNumberConverter(BaseConverter):
```

```
    def to_python(self, value):
```

```
        value = int(value)
```

```
        if value % 2 != 0:
```

```
            raise ValueError()
```

```
        return value
```

```
app = Flask(__name__)
```

```
app.url_map.converters['even'] = EvenNumberConverter
```

```
@app.route('/even/<even:num>')
```

```
def show_even(num):  
    return f"Even number: {num}"
```

📌 URLs:

✓ /even/4 → Even number: 4

✗ /even/3 → 404 (because it's odd)

🧩 In Short — Easy Summary

Converter	Allows Slash /	Example	Output Type
string	✗ No	/user/jani	str
int	✗ No	/num/5	int
float	✗ No	/price/2.5	float
path	✓ Yes	/file/images/photo.jpg	str
uuid	✗ No	/id/UUID_STRING	uuid.UUID

✓ In Simple Words:

- Use **string** for names/text (no /)
 - Use **int** for numbers
 - Use **float** for decimals
 - Use **path** when your data includes /
 - Use **uuid** for unique identifiers (special IDs)
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